MAT 017B Final Exam

September 13, 2018

Name:____

SID:____

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1 \qquad \int \frac{1}{x}\,dx = \ln|x| + C \qquad \int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C \qquad \int \sec^2 x\,dx = \tan x + C$$

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C \qquad \int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int_a^b f(x)\,dx \approx \sum_{k=1}^{n} f\left(x_k\right)\Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x$$

$$\sum_{k=1}^{n} 1 = n \qquad \sum_{k=1}^{n} k = \frac{n\left(n+1\right)}{2} \; \sum_{k=1}^{n} k^2 = \frac{n\left(n+1\right)\left(2n+1\right)}{6}$$

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x) \qquad \frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt = f[g(x)]g'(x) - f[h(x)]h'(x)$$

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C \qquad \int u\,dv = uv - \int v\,du$$

$$A = \int [f(x) - g(x)]\,dx \quad \Delta y = \int \frac{dy}{dx}\,dx \quad \bar{y} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$f\left(x\right) \approx \sum_{k=0}^{n} \frac{f^{(k)}\left(a\right)}{k!}\left(x-a\right)^k \quad \frac{dy}{dx} = f(x)g(y) \quad \int \frac{dy}{g(y)} = \int f(x)\,dx$$

$$y' + p(x)y = q(x) \quad I\left(x\right) = e^{\int p(x)dx} \quad y = \frac{1}{I(x)}\int I(x)q(x)\,dx$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \Delta = \det\left(A\right) = ad - bc \quad \tau = \operatorname{tr}\left(A\right) = a + d$$

$$A\overrightarrow{x} = \lambda\overrightarrow{x} \quad \det\left(A - \lambda I\right) = 0 \quad \lambda^2 - \tau\lambda + \Delta = 0 \quad \left(A - \lambda I\right)\overrightarrow{x} = 0$$

$$|\overrightarrow{x}| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \quad \overrightarrow{x}\cdot\overrightarrow{y} = \sum_{k=1}^{n} x_k y_k = |\overrightarrow{x}||\overrightarrow{y}|\cos\theta \quad \hat{x} = \frac{\overrightarrow{x}}{|\overrightarrow{x}|}$$